

CASE STUDY

Energy Optimisation — Steel Manufacturer

Case Study | Jamshedpur, Jharkhand | 2022

Executive Summary

A large integrated steel plant in Jamshedpur partnered with us to reduce energy expenditure across its hot-rolling, cold-rolling, and furnace operations. By deploying IoT-based energy monitoring and machine-learning predictive analytics, the plant cut energy costs by 18% and avoided approximately **₹8 Cr** in annual overheads within the first year of operation.

18%	₹8 Cr	12 MW	14 mo
Energy Cost Reduction	Annual Savings	Peak Load Shaved	ROI Period

Client Context

Energy accounts for approximately 30% of total operating cost at an integrated steel plant. The client's energy team lacked real-time visibility into consumption patterns and had no predictive model to anticipate peak demand spikes, leading to high Distribution Company (DISCOM) penalties and missed demand-response incentives.

Scope of Work

- IoT sub-metering across 18 major energy consumers (furnaces, rolling mills, compressors)
- Real-time energy dashboard with anomaly alerting and threshold notifications
- ML models for peak-load forecasting (LSTM + XGBoost ensemble) with 96% accuracy
- Demand-response automation: automated load-shedding of non-critical assets during peak windows
- Integration with DISCOM's API for real-time tariff optimisation
- Energy Performance Indicator (EnPI) reporting aligned to ISO 50001

Key Outcomes

- 18% reduction in energy cost per tonne of steel produced
- ₹8 Cr** annual savings — payback achieved in 14 months
- 12 MW peak demand reduction, eliminating DISCOM overdrawal penalties
- Carbon intensity reduced by 14% (Scope 2 emissions), supporting ESG reporting targets
- ISO 50001 certification achieved in Month 16 of programme

Technology Stack

- IoT Edge: Schneider EcoStruxure Power Monitoring Expert with custom MQTT adapters
- Data Platform: Azure Time Series Insights + Azure Data Lake Gen2
- ML Platform: Azure ML (training) + Docker-deployed inference APIs

- Dashboards: Power BI Premium with DirectQuery connections
- Integration: REST API connector to DISCOM demand-response portal